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TRANSPOSITION TECHNIQUES

Changing the position of letters or bits to secure data. Unlike substitution (where characters are replaced), transposition rearranges

RAIL-FENCE CIPHER

•A type of transposition cipher: rearranges letters in plaintext to create ciphertext.

•Named "rail fence" because it resembles the layout of a fence with rails.

•Simple and widely used in classic cryptography.

•In rail fence technique, the plaintext is written down as a sequence of **diagonals** and then read off as a sequence of **rows**.

How Rail Fence Works

- Concept: Write the message in a zigzag (diagonal) pattern across multiple rows.
- After filling the rows, read each row sequentially to get the encrypted message.
- Key here is the number of rows, which defines the cipher pattern.

  For example, to encipher the message “meet me after the toga party” with a rail fence of depth 2, we write the following:

```
m e m a t r h t g p r y
  e t e f e t e o a a t
```

MEMATRHTGPRYETEFETEOAAT

RAIL-FENCE CIPHER

depth 3

Plaintext	T	H	I	S	I	S	A	S	E	C	R	E	T	M	E	S	S	A	G	E
Rail Fence Encoding key = 3	T				I				E				T				S			
		H		S		S		S		C		E		M		S		A		E
			I				A				R				E				G	
Ciphertext	T	I	E	T	S	H	S	S	S	C	E	M	S	A	E	I	A	R	E	G

depth 5

Rail 1	T								G								S						
Rail 2		E						O		I						N		T					
Rail 3			C				L				C				I				I				E
Rail 4				H		O						A		_						T		T	
Rail 5					N								L								U		

Decoding the Rail Fence Cipher

Step 1: Count the letters in each row (based on key rows used in encoding).

Step 2: Arrange letters back into the zigzag pattern.

Step 3: Read diagonally to get the Plaintext.

Example:

Decode "DED ENTE AL FHW L" with 3 rows to get "DEFEND THE WALL".

Practical Activity

Task: Encode “SECURITY CIPHER” using a 3-row rail fence.

Solution Process:

Encoding: Arrange in zigzag with 3 rows, then read row by row.

Decoding: Reverse the steps using the same row count.

Advantages & Limitations

Advantages:

- Simple to understand and implement.
- Hides plaintext pattern slightly.

Limitations:

- Vulnerable to frequency analysis if used alone.
- Easy to decrypt if the row count (key) is known.

Columnar Transposition Cipher

- A **transposition cipher** that rearranges plaintext letters by writing them into columns and reordering the columns based on a key.
- Unlike substitution, it doesn't change the letters, only their order.
- Often used in classical cryptography and serves as a foundational concept for understanding more advanced encryption.

How Columnar Transposition Cipher Works

Step 1: Choose a keyword to determine the column order.

Step 2: Write the plaintext into rows under columns labeled by each letter in the keyword.

Step 3: Rearrange columns in the alphabetical order of the keyword letters.

Step 4: Read columns in this new order to generate the ciphertext.

How Columnar Transposition Cipher Works

Plaintext: "MEET AT DAWN“

Keyword: "3412“

Step 1: Arrange the letters in rows under columns labeled by the keyword:

3	4	1	2
M	E	E	T
A	T	D	A
W	N	_	_

Step 2: Arrange columns in the order of the numbers: 1, 2, 3, 4.

Ciphertext: Read down each column to get "ED_TA_MAWETN".

Decoding the Columnar Transposition Cipher

Step 1: Determine the number of columns and rows from the ciphertext length and keyword.

Step 2: Write down columns in the order dictated by the keyword.

Step 3: Rearrange back into rows to retrieve plaintext.

Example: Decode " ED_TA_MAWETN " using keyword "3412".

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Example 2 - Encoding with a Longer Keyword

Plaintext: “ATTACK AT DAWN”

Keyword: "SECRET“

Step 1: Number the keyword based on alphabetical order

- Alphabetical Order of Letters:"C" is the first letter alphabetically, so it gets 1.
- The first "E" is the next letter alphabetically, so it gets 2.
- The second "E" gets 3.
- "R" is next, so it gets 4.
- "S" is next, so it gets 5.
- "T" is the last, so it gets 6.

  Numbered Keyword:

  S E C R E T

  5 2 1 4 3 6

Encode & Decode Exercise

Activity:

Encode "CONFIDENTIAL INFO" with the keyword "SECURE" using a columnar transposition.

Columnar Transposition Cipher

A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns. The order of the columns then becomes the key to the algorithm. For example,

```
Key:          4 3 1 2 5 6 7
Plaintext:    a t t a c k p
              o s t p o n e
              d u n t i l t
              w o a m x y z
Ciphertext:   TTNAAPTMTSUOAODWCOIXKNLYPETZ
```

Thus, in this example, the key is 4312567. To encrypt, start with the column that is labeled 1, in this case column 3. Write down all the letters in that column. Proceed to column 4, which is labeled 2, then column 2, then column 1, then columns 5, 6, and 7.

Columnar Transposition Cipher

The transposition cipher can be made significantly more secure by performing more than one stage of transposition. The result is a more complex permutation that is not easily reconstructed. Thus, if the foregoing message is re encrypted using the same algorithm,

```
Key:        4 3 1 2 5 6 7
Input:      t t n a a p t
            m t s u o a o
            d w c o i x k
            n l y p e t z
Output:     NSCYAUOPTTWLTMDNAOIEPAXTTOKZ
```

Double Transposition

- The Double Transposition Cipher is a type of encryption method that combines two rounds of transposition ciphers.
- This method involves rearranging the plaintext using two different keys, which adds an additional layer of complexity and security.

Why Use Double Transposition?

A single transposition cipher can be relatively easy to crack, but applying it twice with different keys makes it significantly harder to break

Double Transposition

Choose Two Keys for Columnar Transposition: "KEYWORD1" and "KEYWORD2.“

•These keys will determine the order of the columns in each transposition.

First Transposition (Primary Key): Arrange the plaintext in a grid with columns based on the first keyword (KEYWORD1).

•Use alphabetical order of the letters in the keyword to determine the sequence for reading the columns.

Second Transposition (Secondary Key): Take the intermediate ciphertext and arrange it in a new grid using the second keyword (KEYWORD2).

•Again, use the alphabetical order of the letters in the second keyword to determine the sequence for reading the columns.

•The final output after reading the columns in the order determined by KEYWORD2 is the ciphertext.

Double Transposition

Let’s encrypt "MEET AT DAWN" using the keywords "SECRET" and "ENCRYPT".

Plaintext: "MEET AT DAWN“

Keyword 1: "SECRET“

Keyword 2: "ENCRYPT"

Double Transposition

To visualize the result of this double transposition, designate the letters in the original plaintext message by the numbers designating their position. Thus, with 28 letters in the message, the original sequence of letters is

```
01 02 03 04 05 06 07 08 09 10 11 12 13 14
15 16 17 18 19 20 21 22 23 24 25 26 27 28
```

After the first transposition, we have

```
03 10 17 24 04 11 18 25 02 09 16 23 01 08
15 22 05 12 19 26 06 13 20 27 07 14 21 28
```

which has a somewhat regular structure. But after the second transposition, we have

```
17 09 05 27 24 16 12 07 10 02 22 20 03 25
15 13 04 23 19 14 11 01 26 21 18 08 06 28
```

This is a much less structured permutation and is much more difficult to cryptanalyze.

Thank You

Question and Answers

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